

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

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Finding Reasoning Paths :

- 1 . **Anatomical snuff box** â”Œ **Lunar surface of the dorsal side of the hand** â”Œ **Radial artery** â”Œ **Compound radial artery** .
- 2 . **Vascular injury** â”Œ **Cutaneous abrasion** â”Œ **Finger injury** â”Œ **Arterial injury** â”Œ **Radial artery** .
- 3 . **Finger anatomy** â”Œ **Radial artery anatomical location** â”Œ **Blood flow and bleeding** .

Reasoning Process :

- 1 . **Understanding the clinical scenario**:
 - The patient has a penetrating stab wound in the **anatomical snuff box**, a region located on the **lunar surface of the dorsal side of the hand** .
 - The anatomical snuff box is a classic location for the **radial artery**, which runs through this area and is involved in supplying blood to the dorsal hand and forearm .
- 2 . **Assessing the potential injury**:
 - A stab wound in the anatomical snuff box is highly suggestive of injury to the **radial artery** . This artery is exposed in this region and is a common site for vascular injury .
 - Damage to the radial artery can lead to significant bleeding , as it is a major branch of the brachial artery and provides a substantial blood supply to the hand .
- 3 . **Excluding other arteries**:
 - The **ulnar artery** is located on the medial side of the forearm and is not in the anatomical snuff box . Therefore , it is less likely to be injured in this scenario .
 - The **deep palmar arch** , which is a branch of the radial artery , is located deep in the palm and is not exposed in the anatomical snuff box .
 - Other arteries , such as the dorsal metacarpal arteries or dorsal digital arteries , are minor branches and are unlikely to cause significant bleeding in this context .
- 4 . **Pathophysiology of bleeding**:
 - The radial artery is a compound artery (coiled artery) that makes multiple turns around the wrist . This anatomical configuration makes it particularly vulnerable to injury in the anatomical snuff box .
 - Damage to the radial artery would disrupt its blood flow , leading to rapid and profuse bleeding , which is consistent with the patient’s symptoms .
- 5 . **Clinical relevance**:
 - Immediate management of arterial injury would include surgical intervention , such as direct hemostasis or vascular repair , to control the bleeding .

Conclusion :

The damage most likely responsible for the excessive bleeding is the **radial artery**, a compound branch of the brachial artery . This conclusion is supported by the anatomical location of the injury and the known vulnerability of the radial artery in the anatomical snuff box .

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< answer >

Radial artery

</answer>

